Siting Estee Lu

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Education

PhD **University of Edinburgh**, PhD Economics — Oct 2020 – Aug 2025, Edinburgh, UK
- Thesis Title: Essays on Constrained Decision-Making and Strategic Interactions (Defended, No Corrections: *20 Aug 2025*, Awarded: *1 Sep 2025*)
- Supervisors: Prof. Ed Hopkins & Dr. Axel Gottfries

Visit **Columbia University**, Visiting Scholar — Jan 2025 – May 2025, New York, US
- Cognition and Decision Lab
- Host Faculty: Prof. Michael Woodford

MSc **University of Edinburgh**, MSc Economics (Scottish Graduate Programme in Economics) — Sep 2018 – Aug 2019, Edinburgh, UK
- Dissertation focus on Behavioural Game Theory

BSc **National University of Singapore**, BSc (Honours) Economics — Aug 2012 – Jun 2016, Singapore
- Specialized in Applied and Policy Economics

Research Interests

Behavioural Economics, Game Theory, Learning, Market Design, Large Language Model in Economics

Academic/Teaching Experience

Stipendiary Fellow, University of Edinburgh — Sep 2025 –

2025	ECNM11043 - Econometrics 1 (Postgraduate)
2025	ECNM10070 - Topics in Microeconomics (Year 3 Undergraduate)
2026	ECNM10090 - Experimental Economics (Year 3 Undergraduate)
2026	ECNM10066 - Behavioural Economics (Year 3 Undergraduate)
2025-26	ECNM08013 - Economics 1 (Year 1 Undergraduate)

Tutor, University of Edinburgh — Sep 2020 – Aug 2025

2024	ECNM08016 - Statistical Methods for Economics (Year 2 Undergraduate)
2022-24	ECNM10066 - Behavioural Economics (Year 3 Undergraduate)
2021, 23-24	ECNM08013 - Economics 1 (Year 1 Undergraduate)
2020	ECNM08004 - Economic Principles (Year 1 Undergraduate)

Job Market Paper

Costly Job Search with Inattentive Workers — 2025

Keywords: Rational Inattention, Partially Directed Search

Labour market mismatch can arise from workers having limited attention. This paper proposes a Generalized Partially Directed Search model, extending existing literature by allowing inattentive workers to have diverse priors and heterogeneous attention costs. I show that mismatch can be inherited from bias in workers' default search strategies, and heterogeneous attention costs could contribute to greater variability in equilibrium outcomes. I also explore equilibrium multiplicity. Equilibria where workers adopt different application strategies may generate both higher market efficiency and lower monopsony power than workers employing the same application strategies. This information-theoretic approach offers insights on attention-based policies. Preprint

Publications/Working Papers

Game-theory behaviour of large language models: The case of Keynesian beauty contests — 2025

Economics and Business Review 11 (2), 119-148, 10.18559/ebr.2025.2.2182

Keywords: Large Language Models, Competitive Games

The growing adoption of large language models (LLMs) presents potential for deeper understanding of human behaviours within game theory frameworks. This paper examines strategic interactions among multiple types of LLM-based agents in a classical beauty contest game. LLM-based agents demonstrate varying depth of reasoning that fall within a range of level-0 to 1, which are lower than experimental results conducted with human subjects in previous literature, but they display similar convergence pattern towards Nash Equilibrium choice in repeated setting. Through simulations that varies the group composition of agent types, I found that environment with lower strategic uncertainty enhances convergence for LLM-based agents, and environments with mixed strategic types accelerate convergence for all. Results with simulated agents not only convey insights on potential human behaviours in competitive setting, they also offer valuable understanding of strategic interactions among algorithms.

Learning to Coordinate: Adaptive Learning and Equilibrium Selection in Labour Market 2025

Keywords: Adaptive Learning, Coordination, Job Search

Workers' past application choices can act as heuristics for future decisions. By integrating learning theory into search model, this work explores the role of experiences on workers' application choices. It provides an evolutionary perspective to the labour market dynamics, and offers insights on equilibrium selection. I propose two market structures, where wages are unobservable and observable to workers, and I model workers' application strategies over time using reinforcement learning and best response dynamics respectively. I show that in presence of multiple equilibria, experience-based learning generally leads workers to coordinate on a more efficient, locally asymptotically stable equilibrium, in which they apply with high probability to different firms, in both static and dynamic wage-setting environments. Learning models not only highlight potential mechanisms for equilibrium selection, they also suggest process-oriented policies to improve market efficiency. Preprint

Large Language Models can Predict Human Strategic Decisions *(with Pedro González-Fernández, Helena Normann)* 2026

Keywords: Large Language Models, Strategic Communication, Behaviour Prediction

We study whether LLMs can predict human strategic behaviour from pre-play communication. Using three canonical laboratory games that vary in incentive alignment and communication structure, we provide LLMs and incentivized human forecasters with identical transcripts and ask them to predict players' subsequent actions. GPT-5 consistently outperforms humans and achieves accuracy well above chance, especially when incentives are aligned and communication is bilateral. The performance gap arises almost entirely from correctly forecasting cooperative actions, while both humans and GPT–5 struggle to anticipate defection. These results suggest that strategic communication contains systematic information about future behaviour that humans underutilize and that LLMs are able to exploit more effectively. Preprint

Economics Arena for Large Language Models *(with Shangmin Guo, Haoran Bu, Haochuan Wang, Yi Ren, Dianbo Sui, Yuming Shang)* 2024

Keywords: Large Language Models, Game Theory, Benchmarks

Large language models (LLMs) have been extensively used as the backbones for general-purpose agents, and some economics literature suggest that LLMs are capable of playing various types of economics games. Following these works, to overcome the limitation of evaluating LLMs using static benchmarks, we propose to explore competitive games as an evaluation for LLMs to incorporate multi-players and dynamicise the environment. In this work, we provide an economics arena for the LLMs research community as a dynamic simulation to test abilities of LLMs (i.e. rationality, strategic reasoning ability, instruction-following capability). arXiv

Status Signalling with Luxury and Cultural Goods (MSc Dissertation) 2020

Keywords: Social Status, Signalling, Conspicuous Consumption, Inconspicuous Consumption, Cultural Goods

As the world shifts towards greater consumerism, there is an increasing tendency for individuals to distinguish themselves from the others through the goods that they purchase, which assert the society of their taste and implicit social status. While the pursuit of social status has been well-explored, most studies in this area tend to focus on luxury consumption, limited research investigated the inclusion of cultural goods (books, etiquette classes, etc.) as another signalling tool. The main objective of the paper, therefore, seeks to understand and provide a theoretical grounding for individuals' choices between the status goods, its implications for policy formulations and social mobility. RePEc

Professional Experiences

Fiscal Analyst (Fixed Term), Scottish Fiscal Commission, Edinburgh, UK — Nov 2019 – May 2020
- Property tax, fiscal forecasting, policy briefs, housing price forecasting models.

Market Analyst in Big data, AI/Cognitive Computing Research, IDC Asia Pacific, Singapore — Jan 2018 – Jul 2018
- AI consulting, technology trends and user demand predictions, competitive analysis.

Research Analyst, Datarama Technology, Singapore — Jun 2016 – Jan 2018
- Data sourcing, data architecture design, database management, risk rating

Grants, Awards & Fellowships

2025-26	Moray Endowment Fund, Research Grant (£2,000)
2025	European Economic Association (EEA) Travel Grant
2025	Economics Graduate Research Scholarship
2024	Shortlisted for Student Tutor of the Year (University-wide), Teaching Awards
2023	The Scottish Economic Society Small Grants (£500)
2020-24	Research and Training Scholarship, University of Edinburgh

Conferences/Workshop Presentations

2025	EEA Congress (Bordeaux, France)
2025	15th Annual Conference of the French Association of Experimental Economics (Nancy, France)
2024	Language Gamification Workshop at NeurIPS (Vancouver, Canada)
2024	Foundations of Utility and Risk (FUR) Conference (Brisbane, Australia)
2021-24	Scottish Graduate Programme in Economics (SGPE) PhD Conferences (Edinburgh, UK)
2023	SABE-IAREP Conference (Nice, France)
2023	10th Toulouse Economics and Biology Workshop (Toulouse, France)
2023	5th Queen Mary University of London (QMUL) Economics and Finance Workshop for PhD & Post-doctoral Students, *Presenter and Discussant* (London, UK)
2020	15th Annual International Symposium on Economic Theory, Policy & Applications (Virtual)

Advanced Trainings and Summer Courses

2024	1st Summer School in Experimental and Behavioral Economics, European Economic Review
2024	Cognitive Foundations of Decision-Making Summer School (RISL$\alpha\beta$)
2023	The 33rd Advanced School in Economic Theory: Imperfect Cognition and Economic Behaviour
2023	2nd Summer School of Computational and Experimental Economics (BESLab)
2023	Toulouse Summer School in Quantitative Social Sciences
2023	Replication Games
2022	University of East Anglia Behavioral Game Theory Summer School

Miscellaneous

Activities	Student representative, Programme Rep (PhD Economics), University of Edinburgh
Technical	Python, LaTeX, Mathematica, STATA, SQL, R, Mongodb, Neo4j, IBM i2
Certificates	DELF B1 and B2 French Attestation de Reussite, Financial Risk Management I
Languages	English, Mandarin Chinese, French (Intermediate, B2), German (Basic, A1)
Citizenship	Singapore

References

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